



Mathematical Algorithms for Artificial Intelligence

Course Code : AI255TBC

Semester : 5

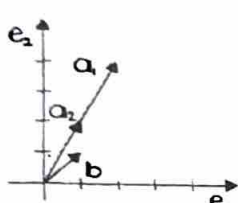
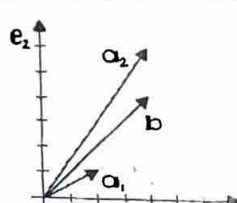
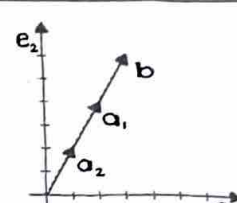
Max Marks : 10+50=60

Date : 27/11/2024

Time : 9:30AM - 11:30AM

Duration : 120 Mins

CIE -1

SL.	Questions	M	B	C
1	Let a_1, a_2 be the two columns of a 2×2 real matrix and b be any other vector in $\mathbb{R}^2$. Consider the figures given below. Identify which of them corresponds to a system of linear equations in two variables with (a) unique solution (b) no solution In each case, justify your choice in not more than 2 sentences.    Fig Q1 (i) Fig Q1 (ii) Fig Q1 (iii)	5	3	2
2.	Which among the following sets of matrices is not a vector space over $\mathbb{R}$, under the usual vector addition and scalar multiplication operations and why? (i) $\mathcal{M}_{2 \times 2}$, the set of real, invertible 2×2 matrices. (ii) $\mathcal{U}_{3 \times 3}$, the set of real, upper triangular matrices	5	2	1
3.	Assert or Reject each of the following 3 statements with proper justification. Statement 1: The set containing a single vector is ALWAYS a linearly DEPENDENT set. Statement 2: The set containing a single vector is ALWAYS a basis for some vector subspace. Statement 3: Every vector space has at least one subspace.	10	3	2
4.	Let A be a 3×3 matrix such that $Ax = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ has both $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$ as its solutions. Find one more solution for the same.	5	3	2
5.	$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$ (i) List out all the four fundamental subspaces associated with the matrix A. (ii) List at least one basis for each of the subspaces that you found above. (iii) Mention the dimension of each of the subspaces alongside.	10	3	2
6.	Let the set $\mathbb{R}^2$ with operations $\oplus$ and $\odot$ defined as follows. For any two vectors $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, in the set, vector addition is defined as $x \oplus y = \begin{pmatrix} x_1 + y_1 - 2 \\ x_2 + y_2 - 3 \end{pmatrix}$ For a real scalar k, scalar multiplication is defined as $k \odot x = \begin{pmatrix} kx_1 - 2k + 2 \\ kx_2 + 3k - 3 \end{pmatrix}$ Find the zero vector $\ominus_2$ for this vector space with the operations as defined above.	5	3	
7.	Consider the following system of equation: $x_1 + x_2 - x_3 + x_4 - x_5 = 0$ Identify the null space associated with the coefficient matrix and its dimension.	5		2
8.	Given the vector space $\mathbb{R}^3$, list out the geometry of its possible subspaces.	5		2

Solution: Linear Systems from CIE-1

Problem Analysis

The problem asks us to relate the solvability of a linear system $Ax = b$ to the geometric arrangement of the column vectors of matrix A (denoted as a_1 and a_2) and the target vector b .

The system of linear equations can be written as a vector equation:

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b}$$

This means we are looking for weights x_1 and x_2 such that the linear combination of vectors a_1 and a_2 equals vector b .

Visual Analysis of the Figures

1. Fig Q1 (i):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line passing through the origin. They are **collinear** (linearly dependent).
- Their span (the set of all possible linear combinations) is just that single line.
- Vector $\mathbf{b}$ points in a different direction and does **not** lie on that line.

2. Fig Q1 (ii):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ point in different directions. They are **non-collinear** (linearly independent).
- Because they are two linearly independent vectors in 2D space ($\mathbb{R}^2$), they form a basis. Their span covers the entire 2D plane.
- Vector $\mathbf{b}$ lies somewhere in that plane (as any 2D vector must).

3. Fig Q1 (iii):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line. They are **collinear** (linearly dependent).
- Vector $\mathbf{b}$ **also** lies on exactly the same line.

Detailed Solutions (Question 1)

(a) Unique Solution

Correct Figure: Fig Q1 (ii)

Mathematical Reasoning: A system $Ax = b$ has a **unique solution** if and only if the columns of A are linearly independent. In $\mathbb{R}^2$, this means the two column vectors must not lie on the same line. In Fig Q1 (ii), $\mathbf{a}_1$ and $\mathbf{a}_2$ span the entire plane, meaning any vector $\mathbf{b}$ can be reached by a unique combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Exam-Style Justification (Max 2 sentences): In Fig Q1 (ii), the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are linearly independent (non-collinear), meaning they span the entire plane $\mathbb{R}^2$. Therefore, any vector $\mathbf{b}$ in the plane can be represented uniquely as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

(b) No Solution

Correct Figure: Fig Q1 (i)

Mathematical Reasoning: A system has **no solution** (is inconsistent) if the target vector $\mathbf{b}$ lies outside the column space (span) of $\mathbf{A}$. In Fig Q1 (i), the span of $\mathbf{a}_1$ and $\mathbf{a}_2$ is a single line. Since $\mathbf{b}$ is off that line, there are no scalar multipliers x_1, x_2 that can make $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$ equal to $\mathbf{b}$.

Exam-Style Justification (Max 2 sentences): In Fig Q1 (i), the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are collinear, so their span is limited to a single line. Since vector $\mathbf{b}$ does not lie on this line, it cannot be generated by any linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Summary for Reference

- **Fig Q1 (i):** No Solution (inconsistent).
- **Fig Q1 (ii):** Unique Solution (consistent and independent).
- **Fig Q1 (iii):** Infinitely Many Solutions (consistent but dependent; $\mathbf{b}$ is on the line spanned by $\mathbf{a}_1, \mathbf{a}_2$).

Solution: Vector Spaces of Matrices (Question 2)

Problem Statement

Identify which of the following sets is **not** a vector space over $\mathbb{R}$: (i) $\mathcal{M}_{2 \times 2}$, the set of real, invertible 2×2 matrices. (ii) $\mathcal{U}_{3 \times 3}$, the set of real, upper triangular matrices.

Detailed Analysis

(i) Set of Invertible 2×2 Matrices ($\mathcal{M}_{2 \times 2}$)

Conclusion: This is **NOT** a vector space.

Reasoning: A vector space must satisfy several axioms, including the existence of an additive identity (the zero vector) and closure under addition.

1. **Missing Zero Vector:** The additive identity in the space of 2×2 matrices is the zero matrix $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. The determinant of the zero matrix is 0, so it is **not invertible**. Therefore, the zero vector is not included in the set.
2. **Not Closed under Addition:** Even if we ignored the zero vector issue, the sum of two invertible matrices is not guaranteed to be invertible.

- Let $A = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (Invertible, $\det = 1$).

- Let $B = -I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ (Invertible, $\det = 1$).
- $A + B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, which is **not** invertible.

Since the set fails to contain the zero vector and is not closed under addition, it is not a vector space.

(ii) Set of Upper Triangular Matrices ($\mathcal{U}_{3 \times 3}$)

Conclusion: This **IS** a vector space.

Reasoning: This set is a subspace of the vector space of all 3×3 real matrices, $M_{3 \times 3}(\mathbb{R})$.

1. **Contains Zero Vector:** The zero matrix consists entirely of zeros. Since all entries below the main diagonal are zero ($0 = 0$), it satisfies the condition of being upper triangular.
2. **Closed under Addition:** If you add two upper triangular matrices, the entries below the diagonal remain zero ($0 + 0 = 0$).
3. **Closed under Scalar Multiplication:** If you multiply an upper triangular matrix by a scalar c , the entries below the diagonal remain zero ($c \times 0 = 0$).

Final Answer

The set that is **not a vector space** is (i) $\mathcal{M}_{2 \times 2}$, the set of real, invertible 2×2 matrices. The primary reason is that it does not contain the zero matrix (additive identity), and it is not closed under vector addition.

Solution: Assertions on Vector Spaces (Question 3)

Problem Statement

Assert or Reject each of the following 3 statements with proper justification.

1. **Statement 1:** The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set.
2. **Statement 2:** The set containing a single vector is **ALWAYS** a basis for some vector subspace.
3. **Statement 3:** Every vector space has at least one subspace.

Detailed Analysis

Statement 1: "The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set."

Verdict: REJECT (False)

Justification: A set containing a single vector $\{\mathbf{v}\}$ is linearly dependent if and only if that vector is the zero vector ($\mathbf{v} = \mathbf{0}$). If the vector is non-zero ($\mathbf{v} \neq \mathbf{0}$), the set is linearly **independent**. This is because the equation $c\mathbf{v} = \mathbf{0}$ implies $c = 0$ when $\mathbf{v}$ is not zero.

- **Counter-example:** Let $V = \mathbb{R}^2$ and let $S = \{(1, 0)\}$. Since $(1, 0) \neq \mathbf{0}$, the set S is linearly independent. Thus, the statement "ALWAYS dependent" is false.

Statement 2: "The set containing a single vector is ALWAYS a basis for some vector subspace."

Verdict: REJECT (False)

Justification: By definition, a basis for a subspace must be a **linearly independent** spanning set. If the single vector in the set is the zero vector, $S = \{\mathbf{0}\}$, then the set is linearly **dependent** (since $1 \cdot \mathbf{0} = \mathbf{0}$, a non-trivial combination yields zero). Therefore, the set $\{\mathbf{0}\}$ cannot be a basis for any subspace. (Note: The subspace generated by the zero vector is the zero subspace $\{\mathbf{0}\}$, but the basis for the zero subspace is conventionally defined as the empty set $\emptyset$, not the set containing the zero vector.)

Statement 3: "Every vector space has at least one subspace."

Verdict: ASSERT (True)

Justification: Every vector space V must contain the additive identity (the zero vector, $\mathbf{0}$). The set $W = \{\mathbf{0}\}$ is a subspace of V because it contains the zero vector and is closed under addition ($\mathbf{0} + \mathbf{0} = \mathbf{0}$) and scalar multiplication ($c \cdot \mathbf{0} = \mathbf{0}$). This is known as the **zero subspace** (or trivial subspace). Therefore, every vector space has at least this one subspace. (Most vector spaces actually have at least two: the zero subspace $\{\mathbf{0}\}$ and the space V itself.)

Solution: Multiple Solutions to a Linear System (Question 4)

Problem Statement

Let A be a 3×3 matrix such that $Ax = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ has both $\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$ as its solutions. Find one more solution for the same.

Detailed Solution

1. Understanding the Linear System Structure

We are given a non-homogeneous linear system $Ax = b$. The general solution to such a system is of the form:

$$x = x_p + x_h$$

where:

- x_p is a particular solution to $Ax = b$.
- x_h is any solution to the homogeneous system $Ax = 0$ (i.e., x_h is in the null space of A).

2. Finding a Homogeneous Solution

Since both $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions to $A\mathbf{x} = \mathbf{b}$, their difference must be a solution to $A\mathbf{x} = \mathbf{0}$.

$$A(\mathbf{x}_1 - \mathbf{x}_2) = A\mathbf{x}_1 - A\mathbf{x}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

Let's calculate this difference vector, let's call it $\mathbf{v}$:

$$\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}$$

So, $A \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}$.

3. Generating a New Solution

We can create a new solution by taking one of the original solutions (like $\mathbf{x}_1$) and adding any scalar multiple of the homogeneous solution $\mathbf{v}$. New Solution $\mathbf{x}_3 = \mathbf{x}_1 + k\mathbf{v}$ (where k is any real number).

Let's choose $k = 1$ for simplicity:

$$\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

Verification

Since $\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{v}$:

$$A\mathbf{x}_3 = A(\mathbf{x}_1 + \mathbf{v}) = A\mathbf{x}_1 + A\mathbf{v} = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

This confirms $\mathbf{x}_3$ is a valid solution.

Alternative Approach (Affine Combination)

Any affine combination of solutions is also a solution (where weights sum to 1).

$\mathbf{x}_3 = 2\mathbf{x}_1 - 1\mathbf{x}_2$ (This matches the calculation above). Another option, $\mathbf{x}_4 = 2\mathbf{x}_2 - \mathbf{x}_1$:

$$2 \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$$

Final Answer

One possible additional solution is:

$$\mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

(Note: There are infinitely many correct answers. Any vector on the line connecting the two given points is a solution.)

Solution: Fundamental Subspaces (Question 5)

Problem Statement

Given matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$, find: (i) The four fundamental subspaces. (ii) A basis for each. (iii)

The dimension of each.

Detailed Analysis

First, we determine the rank of the matrix. The columns are identical, so there is only 1 independent column. **Rank (r) = 1**. Dimensions of matrix A are $m \times n = 3 \times 2$.

- $m = 3$ (rows)
- $n = 2$ (columns)

1. Column Space ($C(A)$)

- **Definition:** The space spanned by the columns of A .
- **Analysis:** Since both columns are the same $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, the span is just the line generated by this single vector.
- **Basis:** $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $r = 1$

2. Row Space ($C(A^T)$)

- **Definition:** The space spanned by the rows of A .
- **Analysis:** All rows are identical $(1, 1)$. The row space is the line in $\mathbb{R}^2$ generated by this vector.
- **Basis:** $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $r = 1$

3. Null Space ($N(A)$)

- **Definition:** The set of vectors x such that $Ax = 0$.

- **Calculation:** $\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ This reduces to the single equation $x_1 + x_2 = 0$, or $x_1 = -x_2$. Vector solution: $x = x_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.
- **Basis:** $\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $n - r = 2 - 1 = 1$

4. Left Null Space ($N(A^T)$)

- **Definition:** The set of vectors y such that $A^T y = 0$.
- **Calculation:** $A^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ Solve $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives one equation: $y_1 + y_2 + y_3 = 0$. We have 2 free variables (y_2, y_3). Let $y_1 = -y_2 - y_3$. Vector solution: $y = \begin{pmatrix} -y_2 - y_3 \\ y_2 \\ y_3 \end{pmatrix} = y_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + y_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.
- **Basis:** $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $m - r = 3 - 1 = 2$

Summary Table

Subspace	Symbol	Dimension	Basis
Column Space	$C(A)$	1	$\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$
Row Space	$C(A^T)$	1	$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$
Null Space	$N(A)$	1	$\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$
Left Null Space	$N(A^T)$	2	$\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$

Solution: Zero Vector in Non-Standard Space (Question 6)

Problem Statement

Find the zero vector Θ_2 for the set $\mathbb{R}^2$ with the following operations: **Vector Addition ($\oplus$)**:

$$\mathbf{x} \oplus \mathbf{y} = \begin{pmatrix} x_1 + y_1 - 2 \\ x_2 + y_2 - 3 \end{pmatrix}$$

Scalar Multiplication ($\odot$):

$$k \odot \mathbf{x} = \begin{pmatrix} kx_1 - 2k + 2 \\ kx_2 + 3k - 3 \end{pmatrix}$$

Detailed Solution

1. Definition of the Zero Vector

The zero vector (additive identity), denoted here as Θ_2 , is the vector such that for any vector $\mathbf{x} \in \mathbb{R}^2$, the following condition holds:

$$\mathbf{x} \oplus \Theta_2 = \mathbf{x}$$

2. Setting up the Equations

Let $\Theta_2 = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$. Substitute $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and Θ_2 into the given addition rule:

$$\mathbf{x} \oplus \Theta_2 = \begin{pmatrix} x_1 + e_1 - 2 \\ x_2 + e_2 - 3 \end{pmatrix}$$

For this result to equal $\mathbf{x}$, each component must match the components of $\mathbf{x}$:

$$\begin{pmatrix} x_1 + e_1 - 2 \\ x_2 + e_2 - 3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3. Solving for Components

We solve for e_1 and e_2 individually:

Row 1:

$$x_1 + e_1 - 2 = x_1$$

$$e_1 - 2 = 0 \implies e_1 = 2$$

Row 2:

$$x_2 + e_2 - 3 = x_2$$

$$e_2 - 3 = 0 \implies e_2 = 3$$

Final Answer

The zero vector Θ_2 for this specific vector space is:

$$\Theta_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Solution: Null Space of a Single Equation (Question 7)

Problem Statement

Consider the following system of equation:

$$x_1 + x_2 - x_3 + x_4 - x_5 = 0$$

Identify the null space associated with the coefficient matrix and its dimension.

Detailed Analysis

1. Identify the Coefficient Matrix (A)

The system consists of a single homogeneous linear equation with 5 variables. The coefficient matrix A is a 1×5 matrix:

$$A = \begin{pmatrix} 1 & 1 & -1 & 1 & -1 \end{pmatrix}$$

We are looking for the Null Space $N(A)$, which is the set of all vectors $\mathbf{x} \in \mathbb{R}^5$ such that $A\mathbf{x} = 0$.

2. Determine Rank and Nullity

- **Number of Variables (n):** 5
- **Rank (r):** The matrix has 1 pivot (the first entry, 1). Thus, $r = 1$.
- **Nullity (Dimension of Null Space):** By the Rank-Nullity Theorem:

$$\text{Nullity} = n - r = 5 - 1 = 4$$

3. Find the Basis for the Null Space

To find the null space, we express the general solution in terms of the free variables.

- **Pivot Variable:** x_1
- **Free Variables:** x_2, x_3, x_4, x_5

Rewrite the equation to solve for the pivot variable:

$$x_1 = -x_2 + x_3 - x_4 + x_5$$

Substitute this into the general vector $\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -x_2 + x_3 - x_4 + x_5 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$$

Decompose this vector into a linear combination of vectors weighted by the free variables:

$$\mathbf{x} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

4. Final Answer

- **Null Space:** The null space is the set of all linear combinations (the span) of the four basis vectors found above.

$$N(A) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

- **Dimension:**

$$\text{Dimension} = 4$$

Solution: Geometry of Subspaces in R3 (Question 8)

Problem Statement

Given the vector space $\mathbb{R}^3$, list out the geometry of its possible subspaces.

Detailed Analysis

Subspaces of $\mathbb{R}^3$ are categorized by their dimension. Since we are working in a 3-dimensional space, the possible dimensions of subspaces range from 0 to 3. Every subspace must pass through the origin (contains the zero vector).

1. Dimension 0: The Origin

- **Algebraic Description:** The set containing only the zero vector $\{\mathbf{0}\}$.
- **Geometry:** A single **point** at the origin $(0, 0, 0)$.
- **Basis:** The empty set $\emptyset$ (or sometimes conventionally denoted as containing the zero vector, though technically independent sets cannot contain zero).

2. Dimension 1: Lines through the Origin

- **Algebraic Description:** The span of a single non-zero vector $\mathbf{v}$. $S = \{c\mathbf{v} \mid c \in \mathbb{R}\}$
- **Geometry:** A **line** passing through the origin.
- **Basis:** Any single non-zero vector lying on that line.

3. Dimension 2: Planes through the Origin

- **Algebraic Description:** The span of two linearly independent vectors $\mathbf{u}$ and $\mathbf{v}$.
 $S = \{c_1\mathbf{u} + c_2\mathbf{v} \mid c_1, c_2 \in \mathbb{R}\}$
- **Geometry:** A **plane** passing through the origin.
- **Basis:** Any two linearly independent vectors lying on that plane.

4. Dimension 3: The Entire Space

- **Algebraic Description:** The entire set $\mathbb{R}^3$.
- **Geometry:** The whole 3D **space** itself.
- **Basis:** Any three linearly independent vectors (e.g., standard basis $\mathbf{i}, \mathbf{j}, \mathbf{k}$).

Summary Table

Dimension	Geometric Object	Restriction
0	Point	Must be the Origin $(0, 0, 0)$